

EDELIVERY SML JA SMP KASUTAMINE EFTI SÜSTEEMIS

Autor: Rihard Rivis

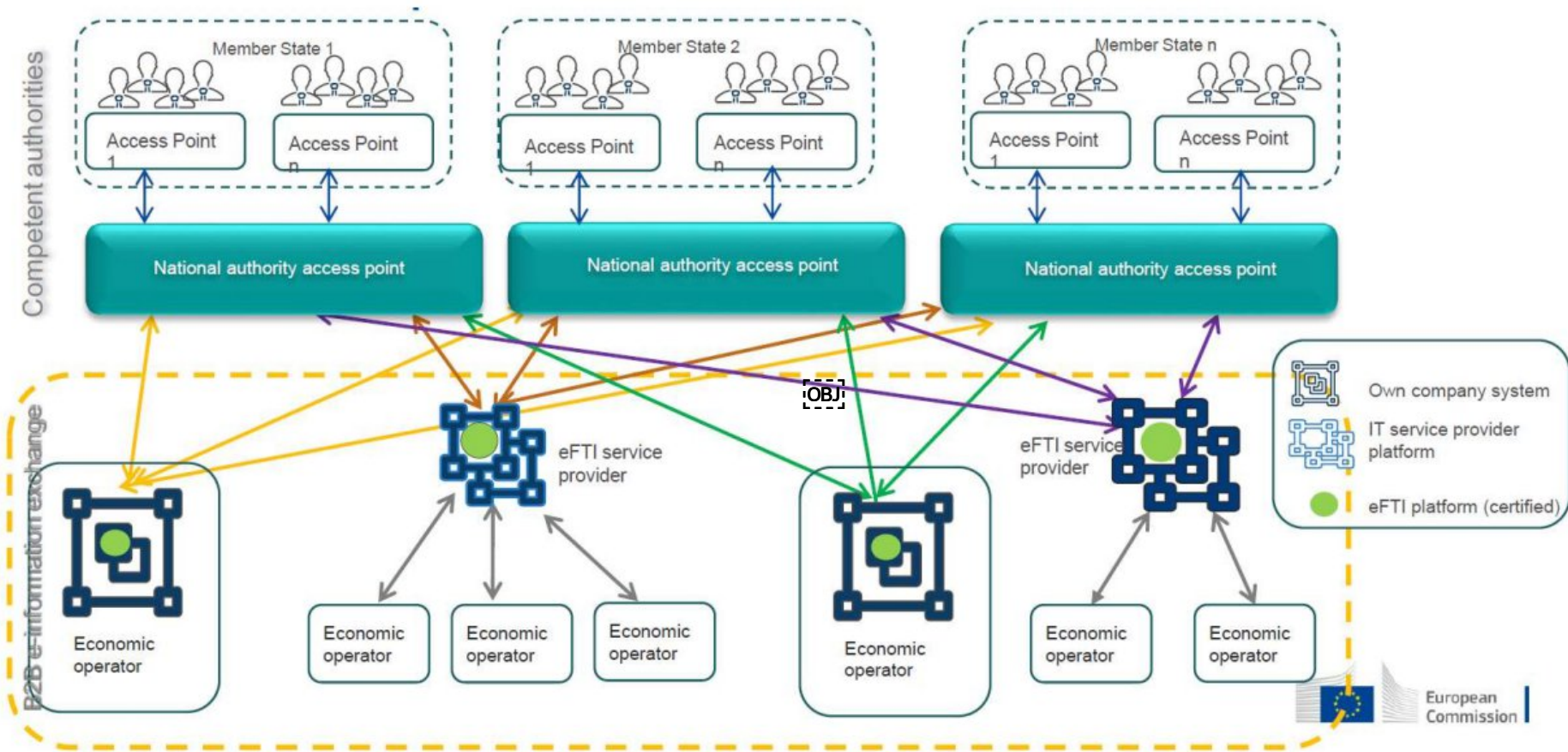
Juhendaja: Ulrika Hurt, Kaasjuhendaja: Tarvo Treier

MIDA SEE TÄHENDAB?

Pealkiri: eDelivery SML ja SMP kasutamine eFTI süsteemis

- eFTI? - EL ülene projekt mille eesmärk on digitaliseerida veoinfo vahetus asutuste vahel.
- eDelivery? - Euroopa liidu standardne andmevahetus lahendus.
- SML/SMP? - Teenuse metaandmete leidja ja otsija.

EFTI ARHITEKTUUR



Allikas: <https://www.dlk.ee/efti-ecosystem/>

PROBLEEM

- eFTI kasutamine muutub EL liikmesriikidele **kohustuslikuks aastal 2027**
- eFTI juurutamiseks on loodud mitu **üleeuroopalist konsortiumit**, mille hulgast eFTI4EU on välja töötanud eFTI näidis arhitektuuri.
- Näidislahendus ei arvesta kõiki eFTI regulatsioonidest tulenevaid nõudeid, näiteks **2024/1942 artikkel 9 (3)**, mis puudutab värava ja platvormi vahelisi päringuid.

EESMÄRK

- **Põhieesmärk:** lahendada eelmisel slaidil toodud probleem.
- **Alameesmärk 1:** olemasolevate valmiskomponentide analüüs.
- **Alameesmärk 2:** näidislahenduse loomine dünaamilise sõnumivahetuse võimaldamiseks.
- **Alameesmärk 3:** jõudluse testimine

ANALÜÜS AP, SMP JA SML KOMPONENTIDELE

Analüüsi käigus osutusid valituks järgmised komponendid:

AP	SMP	SML
Harmony eDelivery access	Phoss SMP server, Harmony SMP server	Ø

NÄIDISLAHENDUSE LOOMINE

Arendatud neljas etapis alustades kõige lihtsamast, töötavast versioonist – staatiline ühendus.

Liikudes samm-sammult täielikult dünaamilise sõnumivahetuseni

ETAPP 1 TULEMUS – STAATILINE KONFIGURATSIOON

Messages

Message Id Message Status From Party Id

To Party Id Message Interval Last 24 hours

Active filters: receivedFrom=2026-04-29 11:19:04 & receivedTo=2026-04-30 11:19:04 & messageType=USER_MESSAGE

Search Reset Search Advanced

10 Show columns

Message Id	From Party Id	To Party Id	Message Status	Received
40452834-8aa4-43c7-9920-14aa2c6fef7f@domibus.eu	sender	receiver	RECEIVED	30-04-2020 14:18:58

0 selected / 1 total

Download Resend Resend All Resend Selected

Saaja(receiver)

Messages

Message Id Message Status From Party Id

To Party Id Message Interval Last 24 hours

Active filters: receivedFrom=2026-04-29 11:33:54 & receivedTo=2026-04-30 11:33:54 & messageType=USER_MESSAGE

Search Reset Search Advanced

10 Show columns

Message Id	From Party Id	To Party Id	Message Status	Received
40452834-8aa4-43c7-9920-14aa2c6fef7f@domibus.eu	sender	receiver	ACKNOWLEDGED	30-04-2020 14:18:54

0 selected / 1 total

Download Resend Resend All Resend Selected

Saatja(sender)

ETAPP 2 TULEMUS – DÜNAAMILINE KONFIGURATSIOON

```
rtid-type:unregistered]]
harmony-ap-sender-1 | 2026-04-30 12:31:26,417 [service_account] [default] [d82b7a98-11cb-45a7-a52b-61e21ee992e6@domibus.eu] [] [http-nio-8080-exec-9] INFO e.d.c.p.p
.d.DynamicDiscoveryPModeProvider:196 - PmodeKey/receiver public certificate not found, starting the dynamic discovery process: [Mandatory field To PartyId is not provid
ed.]
harmony-ap-sender-1 | 2026-04-30 12:31:26,417 [service_account] [default] [d82b7a98-11cb-45a7-a52b-61e21ee992e6@domibus.eu] [] [http-nio-8080-exec-9] INFO e.d.c.p.p
.d.DynamicDiscoveryPModeProvider:246 - Found [1] dynamic discovery candidates: [[eftiProcess]]. MSHRole: [SENDING]
harmony-ap-sender-1 | 2026-04-30 12:31:26,418 [service_account] [default] [d82b7a98-11cb-45a7-a52b-61e21ee992e6@domibus.eu] [] [http-nio-8080-exec-9] INFO e.d.c.p.p
.d.DynamicDiscoveryPModeProvider:626 - Perform lookup by finalRecipient type [null] and value [urn:oasis:names:tc:ebcore:partyid-type:unregistered:C4]
harmony-ap-sender-1 | 2026-04-30 12:31:26,418 [service_account] [default] [d82b7a98-11cb-45a7-a52b-61e21ee992e6@domibus.eu] [] [http-nio-8080-exec-9] INFO e.d.c.p.p
.d.AbstractDynamicDiscoveryService:162 - [SMP] Do the lookup by participant identifier [urn:oasis:names:tc:ebcore:partyid-type:unregistered:C4] and scheme [null], docum
ent identifier [eftiGateAction], process identifier [bdx:noprocess] and scheme [tc1]
harmony-ap-sender-1 | 2026-04-30 12:31:50,446 [service_account] [default] [d82b7a98-11cb-45a7-a52b-61e21ee992e6@domibus.eu] [] [http-nio-8080-exec-9] ERROR e.d.c.p.p
.d.AbstractDynamicDiscoveryService:200 - Could not fetch metadata from SMP for documentId [eftiGateAction] and processId [bdx:noprocess]
harmony-ap-sender-1 | eu.europa.ec.dynamicdiscovery.exception.DNSLookupException: Lookup [CNAME] for participant [ParticipantIdentifier{identifier='c4', scheme='urn:
oasis:names:tc:ebcore:partyid-type:unregistered'} ] has failed. Lookup result CODE [ 2 ]
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.core.locator.dns.impl.DefaultDNSLookup.getAllRecordsForType(DefaultDNSLookup.java:233)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.core.locator.dns.impl.DefaultDNSLookup.dnsRecordNotExists(DefaultDNSLookup.java:166)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.core.locator.impl.DefaultBDXRLocator.cnameLookup(DefaultBDXRLocator.java:160)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.core.locator.impl.DefaultBDXRLocator.getUrlForTopDomain(DefaultBDXRLocator.java:131)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.core.locator.impl.DefaultBDXRLocator.lookupPrivate(DefaultBDXRLocator.java:116)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.core.locator.impl.DefaultBDXRLocator.lookup(DefaultBDXRLocator.java:141)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.service.impl.DynamicDiscoveryService.lookupParticipantSMPUri(DynamicDiscoveryService.java:207)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.service.impl.DynamicDiscoveryService.getDocumentURI(DynamicDiscoveryService.java:193)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.service.impl.DynamicDiscoveryService.getFetcherResponseForServiceMetadata(DynamicDiscoveryService.java:
180)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.service.impl.DynamicDiscoveryService.getServiceMetadata(DynamicDiscoveryService.java:113)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.DynamicDiscovery.getServiceMetadata(DynamicDiscovery.java:52)
harmony-ap-sender-1 | at eu.domibus.core.pmode.provider.dynamicdiscovery.AbstractDynamicDiscoveryService.getServiceMetadata(AbstractDynamicDiscoveryService.ja
va:216)
```


ETAPP 3 TULEMUS – SML SERVERI LOOMINE

```
.d.DynamicDiscoveryPModeProvider:626 - Perform lookup by finalRecipient type [null] and value [urn:oasis:names:tc:ebcore:partyid-type:unregistered:C4]
harmony-ap-sender-1 | 2026-05-03 14:03:46,883 [service_account] [default] [ae26e66e-9e8a-4a60-9a33-aabbafca38c0@domibus.eu] [] [http-nio-8080-exec-7] INFO e.d.c.p.
.d.AbstractDynamicDiscoveryService:162 - [SMP] Do the lookup by participant identifier [urn:oasis:names:tc:ebcore:partyid-type:unregistered:C4] and scheme [null], document identifier [eftiGateAction], process identifier [bdx:noprocess] and scheme [tc1]
harmony-ap-sender-1 | 2026-05-03 14:03:46,884 [service_account] [default] [ae26e66e-9e8a-4a60-9a33-aabbafca38c0@domibus.eu] [] [http-nio-8080-exec-7] INFO e.d.c.p.
omibusProxyServiceImpl:69 - Initialize Domibus proxy.
harmony-ap-sender-1 | 2026-05-03 14:03:46,884 [service_account] [default] [ae26e66e-9e8a-4a60-9a33-aabbafca38c0@domibus.eu] [] [http-nio-8080-exec-7] INFO e.d.c.p.
omibusProxyServiceImpl:72 - Proxy not required. The property domibus.proxy.enabled is not configured
smp | 94115 [Thread-1] INFO SMLServer - Handling query for NAPTR FSPLJSRASNOET76WWBU3VGOATHNBC7HNZ7EUIJXU5LKJFNIZVJJSA.testsmL.
smp | 94335 [4bde-1] INFO RequestLogger - 172.20.0.5 GET /urn:oasis:names:tc:ebcore:partyid-type:unregistered:C4/services/::eftiGateAction: 404 in 8
ms - Apache-HttpClient/5.2.3 (Java/11.0.28)
harmony-ap-sender-1 | 2026-05-03 14:03:47,563 [service_account] [default] [ae26e66e-9e8a-4a60-9a33-aabbafca38c0@domibus.eu] [] [http-nio-8080-exec-7] ERROR e.d.c.p.
.d.AbstractDynamicDiscoveryService:200 - Could not fetch metadata from SMP for documentId [eftiGateAction] and processId [bdx:noprocess]
harmony-ap-sender-1 | eu.europa.ec.dynamicdiscovery.exception.DNSLookupException: SMP lookup address http://smp/urn%3Aoasis%3Anames%3Atc%3Aebcore%3Apartyid-type%3Aunregistered%3AC4/services/%3A%3AeftiGateAction not found - response 404
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.core.fetcher.impl.DefaultURLFetcher.connect(DefaultURLFetcher.java:222)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.core.fetcher.impl.DefaultURLFetcher.fetch(DefaultURLFetcher.java:168)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.service.impl.DynamicDiscoveryService.getFetcherResponseForServiceMetadata(DynamicDiscoveryService.java:182)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.service.impl.DynamicDiscoveryService.getServiceMetadata(DynamicDiscoveryService.java:113)
harmony-ap-sender-1 | at eu.europa.ec.dynamicdiscovery.DynamicDiscovery.getServiceMetadata(DynamicDiscovery.java:52)
harmony-ap-sender-1 | at eu.domibus.core.pmode.provider.dynamicdiscovery.AbstractDynamicDiscoveryService.getServiceMetadata(AbstractDynamicDiscoveryService.java:216)
harmony-ap-sender-1 | at eu.domibus.core.pmode.provider.dynamicdiscovery.AbstractDynamicDiscoveryService.lookupInformation(AbstractDynamicDiscoveryService.java:170)
harmony-ap-sender-1 | at eu.domibus.core.pmode.provider.dynamicdiscovery.AbstractDynamicDiscoveryService$$FastClassBySpringCGLIB$$776ed08b.invoke(<generated>)
harmony-ap-sender-1 | at org.springframework.cglib.proxy.MethodProxy.invoke(MethodProxy.java:218)
harmony-ap-sender-1 | at org.springframework.aop.framework.CglibAopProxy$CglibMethodInvocation.invokeJoinpoint(CglibAopProxy.java:792)
```

ETAPP 4 TULEMUS – SMP SERVERI LOOMINE

localhost:8091

Messages

Message Id

Message Status

From Party Id

To Party Id

Message Interval

Last 24 hours

Active filters: receivedFrom=2026-05-08 08:45:00 & receivedTo=2026-05-09 08:45:00 & messageType=USER_MESSAGE

Search

Reset Search

Advanced

10

Show columns

Message Id	From Party Id	To Party Id	Message Status	Received
4b9ddabd-f2f1-4470-99bd-0640ba1517c0@domibus.eu	sender	receiver	RECEIVED	09-05-2026 11:44:28

localhost:8090

Messages

Message Id

Message Status

From Party Id

To Party Id

Message Interval

Last 24 hours

Active filters: receivedFrom=2026-05-08 08:44:47 & receivedTo=2026-05-09 08:44:47 & messageType=USER_MESSAGE

Search

Reset Search

Advanced

10

Show columns

Message Id	From Party Id	To Party Id	Message Status	Received
4b9ddabd-f2f1-4470-99bd-0640ba1517c0@domibus.eu	sender	receiver	ACKNOWLEDGED	09-05-2026 11:44:27

Terminal - edelivery-smp-sml

Terminal docker x Local x Local (2) x + AI Agents

```
harmony-ap-sender-1 | 2026-05-09 08:44:26,535 [service_account] [default] [4b9ddabd-f2f1-4470-99bd-0640ba1517c0@domibus.eu] [] [http-nio-8080-exec-9] INFO e.d.c.p.Dom
ibusProxyServiceImpl:69 - Initialize Domibus proxy.
harmony-ap-sender-1 | 2026-05-09 08:44:26,536 [service_account] [default] [4b9ddabd-f2f1-4470-99bd-0640ba1517c0@domibus.eu] [] [http-nio-8080-exec-9] INFO e.d.c.p.Dom
ibusProxyServiceImpl:72 - Proxy not required. The property domibus.proxy.enabled is not configured
smp | 161073 [Thread-1] INFO SMLServer - Handling query for NAPTR SG7WQEA6Z27EYWS575WTM3AZ6SVFZ1Y7G6TRTKZDWC45AZRV60EQ.testsm1.
smp | 161200 [dc98-1] INFO RequestLogger - 172.20.0.3 GET /urn:oasis:names:tc:ebcore:partyid-type:unregistered::receiver/services/::eftiGateAction: 200
in 27 ms - Apache-HttpClient/5.2.3 (Java/11.0.28)
harmony-ap-sender-1 | 2026-05-09 08:44:26,883 [service_account] [default] [4b9ddabd-f2f1-4470-99bd-0640ba1517c0@domibus.eu] [] [http-nio-8080-exec-9] INFO e.d.c.c.Cer
tificateServiceImpl:578 - Adding certificates [[receiver]] to [/var/opt/harmony-ap/etc/ap-truststore.p12]
harmony-ap-sender-1 | 2026-05-09 08:44:26,886 [service_account] [default] [4b9ddabd-f2f1-4470-99bd-0640ba1517c0@domibus.eu] [] [http-nio-8080-exec-9] INFO e.d.c.c.Cer
tificateServiceImpl:640 - Added certificate with alias [receiver]: [[
harmony-ap-sender-1 | [
harmony-ap-sender-1 | Version: V3
harmony-ap-sender-1 | Subject: CN=receiver
harmony-ap-sender-1 | Signature Algorithm: SHA256withRSA, OID = 1.2.840.113549.1.1.11
harmony-ap-sender-1 | Key: Sun RSA public key, 2048 bits
harmony-ap-sender-1 | params: null
harmony-ap-sender-1 | modulus: 25433231260493465416124611009854462964936934870452059440899640777140898469357620471471013763679438102826372865998719622512652770956965
```

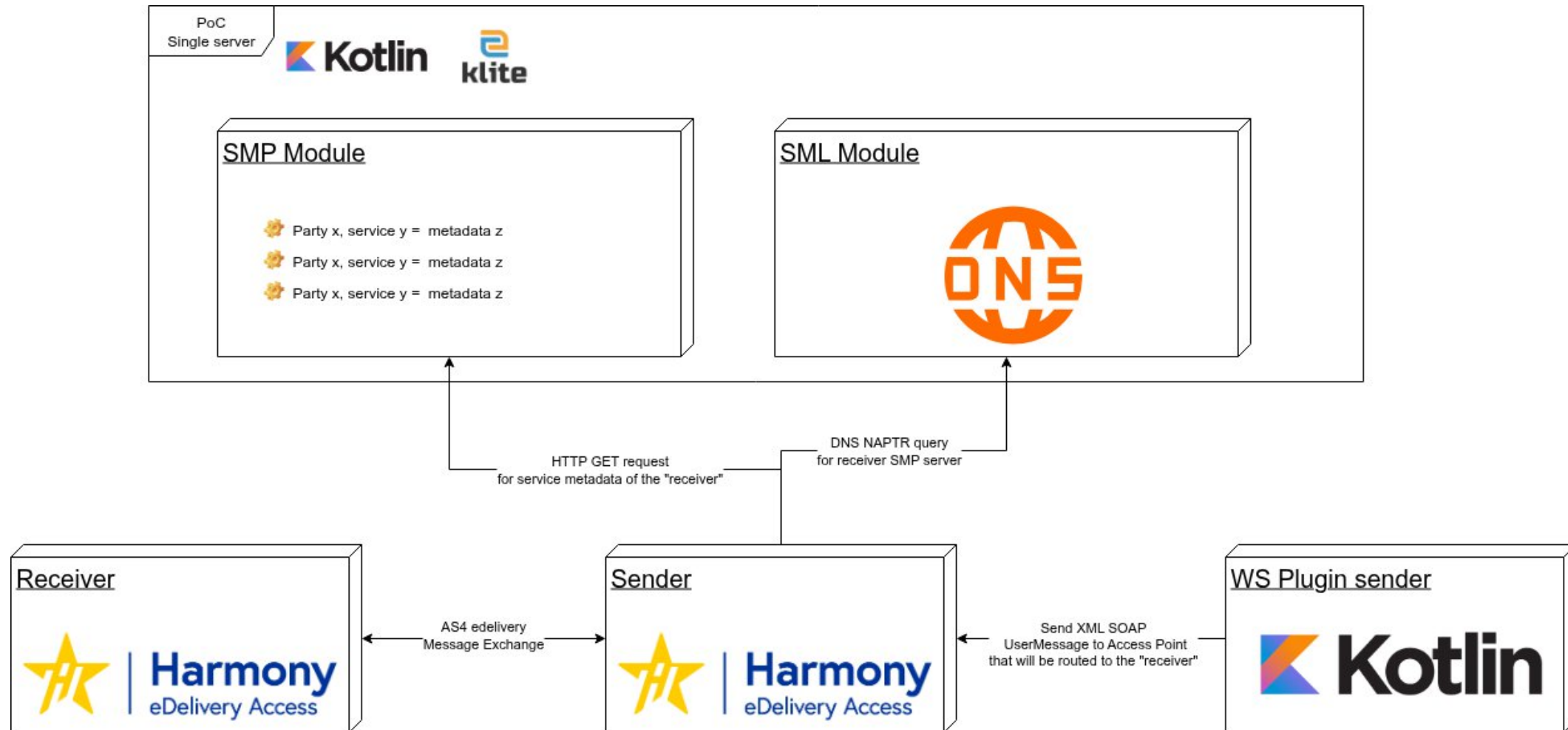
Download

Resend

Resend All

Resend Selected

NÄIDISLAHENDUSE ARHITEKTUUR



JÕUDLUSTESTIDE TULEMUSED

Koormus		Päringuid sekundis		
Päringute arv (-n)	Samaaegsed päringud (-c)	POC SMP	Phoss SMP	Harmony SMP
100	10	3134	1245	132
1000	100	920	3080	178
5000	250	3220	1297	209
10000	1000	4920	1290	211
Koormus		99% latentsus (ms)		
Päringute arv (-n)	Samaaegsed päringud (-c)	POC SMP	Phoss SMP	Harmony SMP
100	10	3.2	8	75
1000	100	108	32	559
5000	250	77	192	1191
10000	1000	200	774	4725

The logo for TAL TECH, featuring the words "TAL" and "TECH" stacked vertically in a bold, white, sans-serif font. The background is a solid magenta color.

**TAL
TECH**

TALLINNA TEHNIKAÜLIKOO

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